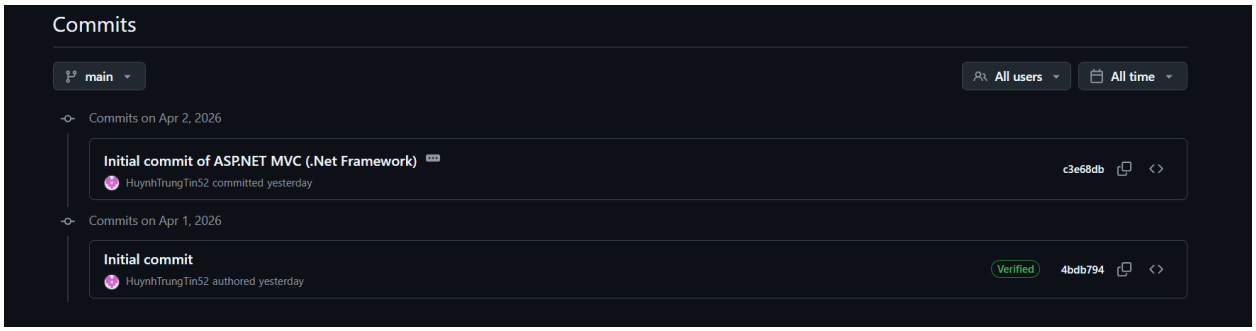
SE Lab 6: ASP.NET MVC Summary Report

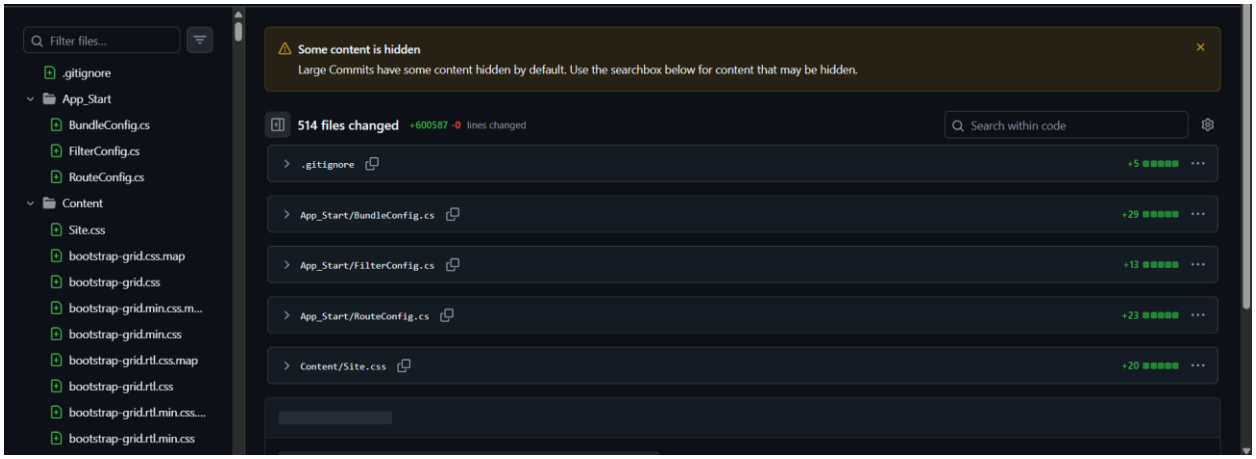
Introduction : The purpose of these exercises was to learn how to build dynamic web applications using ASP.NET MVC (.NET Framework) and the Database First approach. The lab focused on connecting to an MSSQL database using Entity Framework, implementing CRUD operations, managing source code with GitHub, and extending the application with authentication and reporting features.

Exercises

Exercise 1: Setting Up GitHub Repository for ASP.NET MVC

- **Description:** Initialized a local Git repository for the ASP.NET MVC project and mapped it into a remote GitHub repository (ASPNETWebAppMVCStudentApp) for version control.
- **What I learned:** How to manage .NET source code and commit/push changes directly from Visual Studio.

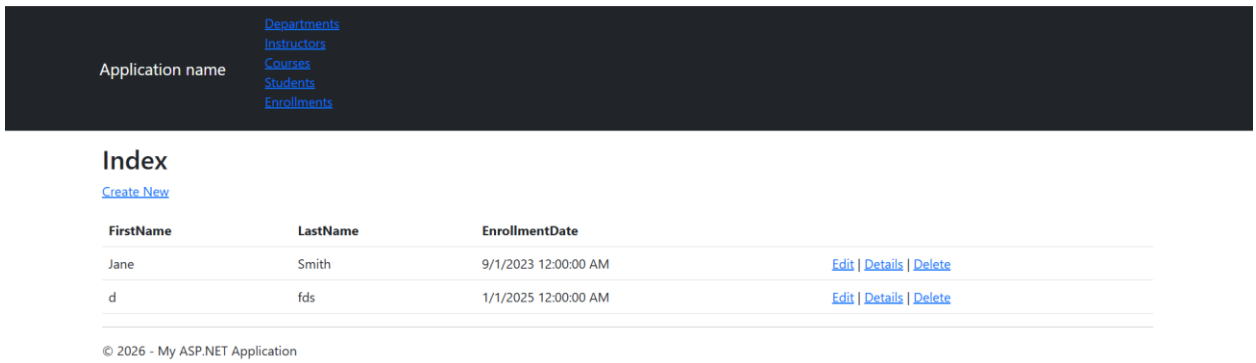


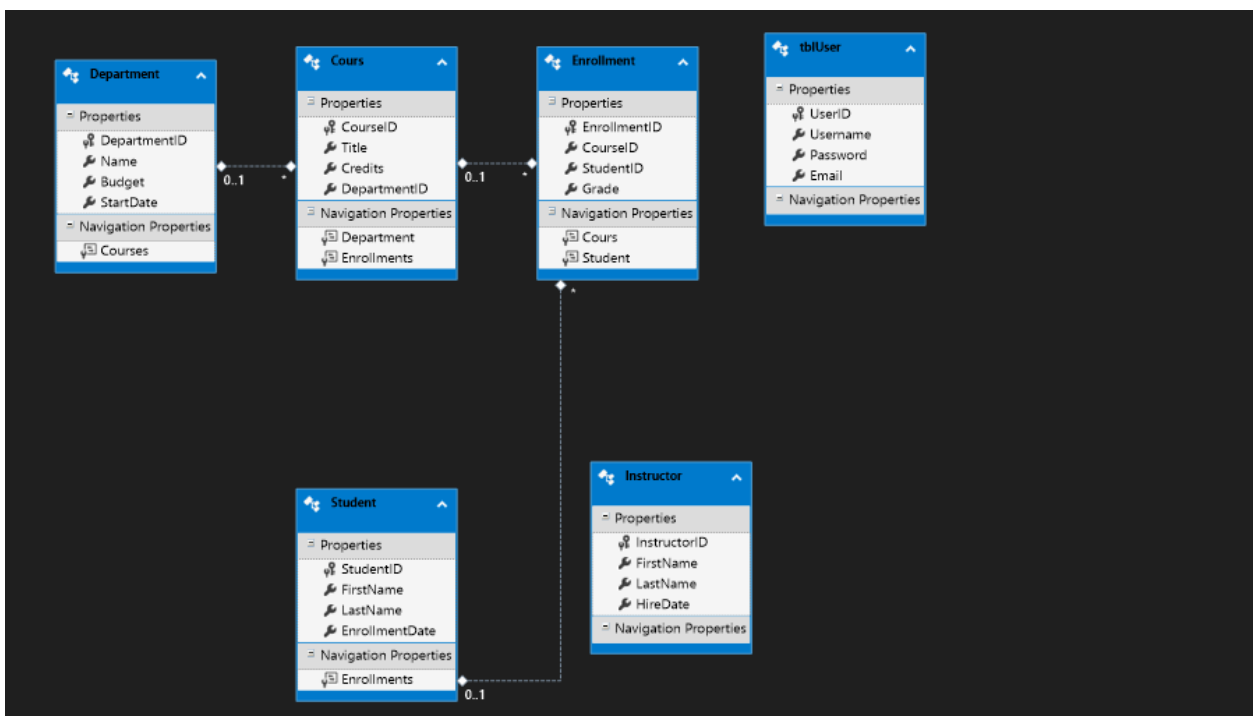


Link: https://github.com/HuynhTrungTin52/ASPNETMVCStudent

Exercise 2: School Management Web App with ASP.NET MVC

- **Description:** Created the SchoolDB database and generated an Entity Framework model (Database First). Scaffolding was used to quickly generate controllers and views for Students, Courses, Instructors, Enrollments, and Departments.
- **What I learned:** How to implement full CRUD (Create, Read, Update, Delete) functionality and modify layout navigation menus in an MVC architecture.



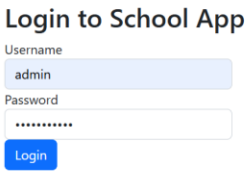


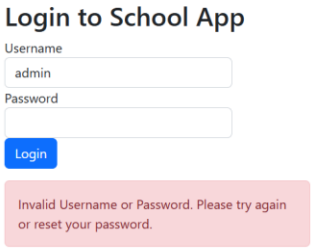
(The tblUser is for the next exercise)

Link: https://github.com/HuynhTrungTin52/ASPNETWebAppMVCStudentApp

Exercise 3: Challenge to get credit points (User Authentication)

- **Description:** Enhanced the School App by adding a tblUser table to the database and creating a secure login system. Configured the application route to display the login form as the very first screen.
- **What I learned:** How to validate user credentials against a database, manage user sessions, and alter default routing in RouteConfig.cs.





Exercise 4: Report in ASP.NET MVC

- **Description:** Installed the RDLC Report Viewer package and designed a visual report (StudentReport.rdlc) bound to a custom query joining Students and Enrollments. Unable to finish the views due to the lack of time and problem with installing the extension for reports management
- **What I learned:** How to bypass older template issues to integrate Microsoft Reporting tools into a modern MVC application

[I would add a picture here but VS has crashed 5 times when I tried to open up the report, will update this later on]

Conclusion

Overall, this lab provided comprehensive, hands-on experience with the ASP.NET MVC framework. The biggest challenges were configuring the third-party RDLC reporting tools and manually overriding the missing report templates. However, successfully implementing these features, alongside the custom user authentication system, greatly improved my understanding of how data flows between the database, controllers, and views in a full-stack .NET application.